

Deepak Mallampati

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US-based, F-1 OPT, open to sponsorship

PROFESSIONAL SUMMARY

Applied AI engineer (2.5+ years) who ships customer-facing agentic systems end-to-end: multi-provider LLM abstractions, schema-validated agent handoffs, RAG-grounded workflows, and FastAPI/Docker packaging in HIPAA-conscious environments. Solo shipper of Manga Lens on the Chrome Web Store (4 AI vision providers behind one abstraction, 29 sites) and founder of E-Farming (80-120 active users). Master's degree, Kent State (May 2025). Open to relocating to San Francisco and frequent customer-site travel. US-based, F-1 OPT, open to sponsorship.

CORE COMPETENCIES

LLM SDK Integration (Claude, GPT-4o, Ollama)

Agentic Workflows + Tool Discipline

RAG + Vector Search + Grounding

FastAPI / Flask / Docker

Python + TypeScript + React

REST/JSON + Multi-Provider Abstraction

Production ML (scikit-learn, XGBoost)

End-to-End Delivery (founder, solo ship)

PROFESSIONAL EXPERIENCE

Jr. AI/ML Engineer Trainee - Progress Solutions Inc.

Jul 2025 - Present, USA - Healthcare Technology, AI Consulting, SaaS

Built **agentic LLM workflows** for multi-step healthcare queries (eligibility checks, care workflow navigation, documentation clarification) with structured reasoning, tool discipline, and grounding rules; improved agent response stability by **~25%**.

Built **Retrieval-Augmented Generation (RAG) systems** for clinical knowledge retrieval; recursive semantic chunking + transformer embeddings; improved contextual retrieval precision by **~35%** and reduced irrelevant retrieval by **>30%**.

Retrieval-grounded response alignment exceeded **90%**.

Packaged LLM/ML inference as **FastAPI/Flask RESTful services**, containerized with **Docker**, structured logging and load simulation; reduced post-deployment defects by **~30%**.

Shipped **predictive ML pipelines** (scikit-learn, XGBoost) for no-show prediction and care engagement scoring; improved recall by **15-20%** on high-risk categories while holding precision **>90%**.

Maintained HIPAA-conscious data governance: de-identification, lineage docs, evaluation audit trails, stakeholder-facing system-limitation docs.

Founder & Full-Stack Developer - E-Farming Digital Marketplace

Independent project - AgriTech, two-sided marketplace

Independently designed and shipped a **PHP/MySQL/Bootstrap marketplace** connecting small-scale farmers with buyers.

Onboarded **80-120 active users** in phase one; cart, reviews, and community blogging features to drive repeat engagement.

Owned discovery-to-deployment loop with no playbook.

Database & DevOps Performance Engineer (Intern) - Energy Solutions International (Emerson)

Jun 2022 - Apr 2023, Hyderabad - Enterprise ERP, Industrial Automation

Optimized SQL and T-SQL stored procedures powering the Synthesis Order-to-Cash platform; reduced query execution time **~20%** and data retrieval latency **~25%**.

Built **CI/CD pipelines with Jenkins** for schema updates and stored-procedure deployments; reduced deployment errors **>30%** and improved release cycle time **~35-40%**.

Machine Learning Engineer - Suvidha Foundation

Jan 2022 - Mar 2022, Hyderabad - Nonprofit Tech, Applied NLP

Built a **transformer-based video summarization and highlight system** across 5,000+ sessions; manual review time reduced **60-70%**; **~85% alignment** with human curation. Deployed as a Flask API with a lightweight UI for non-technical staff.

SELECTED PROJECTS

Manga Lens - *Chrome Web Store extension, solo.* Manifest V3 + content scripts + service workers + viewport-slice capture; **4 AI vision providers** (Claude Sonnet, GPT-4o mini, GPT-4.1 Nano, Ollama/minicpm-v) behind one abstraction; per-domain selector configs for **29 sites**; per-provider payload handling (WebP cloud, JPEG Ollama); 7-day translation cache; privacy policy + narrowed host permissions.

Agentic Healthcare Claims Processing & Fraud Risk Intelligence System. Multi-agent pipeline (Intake -> Validation -> Consistency -> Duplicate Detection -> Fraud Risk Scoring) with **schema-validated JSON contracts** between agents to prevent cascading hallucinations. RAG-grounded CPT/ICD validation against vector-indexed policy documents; ANN duplicate detection; explainable risk scoring with audit-ready reasoning traces.

Dream Decoder - Full-stack *FastAPI + React/TypeScript/Vite/Tailwind* multimodal app coordinating Perplexity Sonar + Replicate behind one orchestrator. Intermediate structured prompt transformation layers raised contextual alignment **~30%** and first-pass image success **~25-30%**.

Agentic Pixel Character Synthesis Engine (*ongoing*). Stable Diffusion + LoRA + ControlNet + OpenPose/MediaPipe + latent-space consistency; **LLM-based orchestrator** decomposes high-level prompts into generation tasks. Backend: PyTorch, Diffusers, FastAPI, Docker.

EDUCATION

Master's Degree - *Kent State University, Ohio, USA (Aug 2023 - May 2025)*

SKILLS

Languages & Frameworks: Python, FastAPI, Flask, TypeScript, React, SQL (T-SQL, PostgreSQL, SQLite), C++ (Arduino), PHP

LLM & GenAI: LLM SDK integration, RAG, agentic workflows, structured outputs, prompt engineering, evaluation pipelines, guardrails, grounding, semantic retrieval, embeddings, vector search

ML Libraries: scikit-learn, XGBoost, PyTorch, Hugging Face Transformers, Diffusers, LangChain, LlamaIndex

Multimodal & CV: YOLOv8, OpenCV, ControlNet, OpenPose/MediaPipe, Stable Diffusion

Infra & DevOps: Docker, Jenkins, Grafana, CI/CD, MLOps/LLMOps, RESTful APIs, observability/logging, REST/JSON

Domains: Applied AI, Healthcare AI (HIPAA-conscious), Workflow Automation, Multimodal Systems